

Topic 1: TCP vs UDP

TCP establishes a connection and delivers a reliable, ordered byte stream. TCP can retransmit lost data, and a receiving application reads bytes in order rather than receiving separate message boundaries.

UDP sends individual datagrams without establishing a TCP-style connection. UDP does not guarantee delivery or ordering, so an application must handle loss or reordering if that matters.

A synthetic file-copy application needs every byte in order, so TCP fits that requirement. A synthetic position-update application can discard an old update and use the newest one, so it can choose UDP when occasional loss is acceptable. These examples describe different requirements; UDP is not a guarantee of faster delivery.

Topic 2: Ports

TCP and UDP port numbers identify communication endpoints for applications on a host. A port number is a 16-bit value and ranges from 0 to 65535. An IP address selects a host interface; the port identifies the application endpoint used for the conversation.

In these synthetic service examples, an SSH server listens on TCP port 22, an IMAP server listens on TCP port 143, and a PostgreSQL server listens on TCP port 5432. A client selects the server's listening port to contact that service.

A TCP endpoint and a UDP endpoint with the same numeric port are distinct. A connection is distinguished by its protocol and the source and destination addresses and ports. Two clients can contact the same server port using different source ports.

Topic 3: Subnetting

Subnetting divides an address block into smaller networks. An IPv4 /24 prefix uses 24 network bits and leaves 8 host bits. The subnet mask for /24 is 255.255.255.0.

The synthetic subnet 198.51.100.0/24 has 256 total addresses. Its network address is 198.51.100.0 and its broadcast address is 198.51.100.255. In a conventional /24 subnet, the host range is 198.51.100.1 through 198.51.100.254.

Splitting 198.51.100.0/24 into two /25 subnets yields 198.51.100.0/25 and 198.51.100.128/25. Each /25 uses 25 network bits, leaves 7 host bits, and contains 128 total addresses. Its subnet mask is 255.255.255.128.